

Functional Annotation Report: UreF

(Urease accessory protein UreF)

Gene: *ureF* — **UniProt:** Q88J01 — **Ordered locus:** PP_2848 **Organism:** *Pseudomonas putida* KT2440 (strain ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950) **Protein family:** UreF family (Pfam PF01730; InterPro IPR002639 / IPR038277 "UreF_sf"); HAMAP rule MF_01385 **Length:** 224 aa

0. Identity verification (mandatory)

The target is unambiguous. The gene symbol *ureF*, the UniProt description ("Urease accessory protein UreF"), the UreF protein family/domains (PF01730, IPR002639), and the organism (*P. putida* KT2440) are mutually consistent. Genome mapping places *ureF* (PP_2848) inside a complete, contiguous chromosomal urease (*ure*) gene cluster (PP_2842–PP_2849) that contains all three urease structural subunits and all four canonical accessory proteins (see §4). Extensive mechanistic literature exists for UreF orthologs in *Klebsiella aerogenes* and *Helicobacter pylori*; those studies concern the **same** protein family and are directly transferable. **No conflicting "same-symbol, different-gene" literature was encountered.** The *P. putida* protein itself is characterized largely by curated database inference and cross-organism homology rather than by dedicated primary studies on the KT2440 protein.

1. Summary (answer to the research question)

UreF is **not an enzyme, transporter, or signaling molecule**; it is a **cytoplasmic molecular chaperone / assembly scaffold** dedicated to the post-translational maturation of the nickel metalloenzyme **urease** (urea amidohydrolase, EC 3.5.1.5). UreF's primary molecular function is to bind the accessory protein UreD (UreH) on the urease apoprotein and to **recruit and activate the GTPase UreG**, assembling the (UreD–UreF–UreG)–apourease "pre-activation complex" through which nickel is inserted into the urease active site in a GTP-hydrolysis-dependent manner. UreF does not catalyze urea hydrolysis itself; its output is a

correctly matured, Ni-loaded urease that in turn hydrolyzes urea to ammonia and carbamate/CO₂, enabling the cell to use urea as a nitrogen source. UreF functions entirely in the cytoplasm.

2. Primary function: what UreF does at the molecular level

2.1 Role class: accessory/maturation chaperone (not catalytic)

Urease is a dinuclear-nickel enzyme that hydrolyzes urea to ammonia and carbamate, the latter decomposing to a second ammonia plus carbonic acid (urease = urea amidohydrolase, EC 3.5.1.5) [PMID:7565414]. The catalytic active site resides in the large structural subunit (UreC/ α in *P. putida*, PP_2845), which carries the dinuclear Ni center coordinated by a carbamylated lysine. Assembling that active site requires the incorporation of **both CO₂ (as a carbamylated Lys ligand) and two Ni²⁺ ions**, a process that most ureases cannot accomplish spontaneously and instead achieve with the help of accessory proteins UreD (UreH), UreE, UreF, and UreG [PMID:7565414;

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Within this system UreF is a soluble accessory protein with **no urea-hydrolysis activity of its own**. The curated record for the *P. putida* protein states its function directly: "Required for maturation of urease via the functional incorporation of the urease nickel metallocenter," and describes UreD, UreF, and UreG as forming "a complex that acts as a GTP-hydrolysis-dependent molecular chaperone, activating the urease apoprotein by helping to assemble the nickel containing metallocenter of UreC" (UniProt Q88J01; HAMAP MF_01385). Reviews of urease metallocenter biosynthesis reach the same conclusion: nickel is inserted "in a GTP-dependent process with the assistance of UreD/UreH, UreE, UreF, and UreG... orchestrat[ing] apoprotein activation by delivering the appropriate metal, facilitating protein conformational changes, and possibly providing a requisite post-translational modification" [PMID:23539618].

2.2 Scaffold that recruits the UreG GTPase

The defining molecular action of UreF is to bridge UreD/UreH (which is bound to the urease apoprotein) and the GTPase UreG, thereby nucleating the activation complex. Structural work on the *H. pylori* orthologs makes this concrete:

- The **UreF–UreH crystal structure** (a dimer of heterodimers, ~116 kDa) shows that complex formation orders UreF's flexible C-terminus into an extra helix (α 10) stabilized by Arg-250 and uncoils the first turn of helix α 2 to expose the conserved residue **Tyr-48**. The single-residue substitutions **R250A** or **Y48A** abolish formation of the UreG–UreF–UreH complex and abolish urease maturation, demonstrating that UreF's C-terminal region and α 2 are essential for **recruiting UreG** to build the pre-activation complex [PMID:22013070].
- Mapping of conserved UreF surface residues in *K. aerogenes* (16 Ala substitutions) defines a discrete **UreG-binding surface**: variants that weaken UreF–UreG interaction are precisely the ones that lose urease activity [PMID:22369361].
- UreF adopts an **all- α -helical fold** and forms a homodimer, with evolutionary-conservation analysis highlighting a probable protein–protein interaction surface [PMID:20635345].

2.3 Structurally-resolved mechanism: UreG dimerization and interfacial Ni-site assembly

The crystal structure of the ternary **UreG/UreF/UreH complex** provides the step-resolved mechanism [PMID:24115911]:

1. UreF and UreH form a complex that **recruits UreG** (a SIMIBI-class GTPase) to the pre-activation complex.
2. UreF/UreH **facilitate dimerization of UreG** and, in doing so, **juxtapose two invariant Cys66-Pro67-His68 metal-binding motifs** at the UreG dimer interface, assembling a nickel-binding site within a (UreG/UreF/UreH)₂ assembly.
3. Adding **Ni²⁺ and GTP releases a nickel-charged UreG dimer** that binds one Ni ion at the dimer interface; C66A/H68A substitutions abolish formation of this nickel-charged dimer.
4. Upon **GTP hydrolysis**, the UreG dimer reverts to a monomer and **releases nickel to urease**.

Thus UreF's precise contribution is to template the productive quaternary arrangement of UreG so that UreG becomes a competent, GTP-gated nickel donor. A review of the pathway integrates this with the upstream steps: UreE delivers nickel to UreG (GTP promotes a UreE₂G₂ complex in which UreG receives Ni from UreE); Ni/GTP-bound UreG then joins UreF/UreH/apo-urease, and GTP hydrolysis releases Ni into the urease active site [PMID:35556134].

2.4 Nickel-transfer tunnel and fidelity ("quality control")

Two additional lines of evidence sharpen UreF's role:

- **Directional, protected metal delivery.** In silico analysis of the *H. pylori* UreH/UreF/UreG complex identified a water tunnel originating at a nickel-binding motif in UreG, **passing through UreF**, and exiting UreD/UreH toward the urease apoprotein. In *K. aerogenes*, engineered variants that disrupt a ~34.6 Å internal tunnel greatly reduced urease specific activity and lowered nickel content while **increasing Zn/Fe misincorporation**, without disrupting protein–protein interactions; molecular dynamics confirmed tunnel narrowing in these variants [PMID:26401965]. UreF therefore forms part of a physical conduit for sequestered "hand-to-hand" nickel transfer, which enforces **metal selectivity (Ni over Zn/Fe)** and avoids releasing toxic free Ni²⁺.
- **Timing/fidelity control.** UreD–UreF–urease apoprotein complexes are resistant to inactivation by NiCl₂ and require less bicarbonate for activation than urease or UreD–urease complexes; UreF masks UreD in the complex. This led to the proposal that UreF binding **excludes nickel from the active site until after the carbamylated-lysine ligand has formed** [PMID:8808930]. UreF was further proposed to act like a **GTPase-activating protein (GAP)** coupling UreG GTP hydrolysis to productive activation: a K165A UreF variant **uncouples** elevated GTPase activity from urease activation, indicating that wild-type UreF enhances the **fidelity** with which GTP hydrolysis is converted into a correctly metalated enzyme [PMID:22369361].

Substrate/ligand specificity note. UreF has no small-molecule substrate. Its functional "specificity" is for its protein partners (UreD/UreH and UreG) and, through the complex it builds, for delivering Ni²⁺ (not Zn²⁺/Fe²⁺) to urease.

3. Subcellular localization

UreF acts in the **cytoplasm**. The curated record for Q88J01 assigns "Subcellular location: Cytoplasm" and the keywords "Chaperone, Cytoplasm, Nickel insertion." This is consistent with the whole urease system being soluble/cytoplasmic [PMID:7565414] and with all mechanistic complexes (UreD–UreF, UreF–UreH, UreG/UreF/UreH, and the pre-activation complex with apo-urease) being characterized as soluble assemblies. In the *P. putida* KT2440 cluster, every Ure protein is annotated cytoplasmic except the integral-membrane accessory protein UreJ (see §4).

4. Genomic and pathway context in *P. putida* KT2440

ureF (PP_2848) lies within a complete, contiguous chromosomal *ure* cluster (ordered loci PP_2842–PP_2849). Mapping the annotated members:

Locus	Gene	Product	Length	Localization
PP_2842	<i>ureD</i>	Urease accessory protein UreD	277	Cytoplasm
PP_2843	<i>ureA</i>	Urease subunit gamma (γ)	100	Cytoplasm
PP_2844	<i>ureB</i>	Urease subunit beta (β)	105	Cytoplasm
PP_2845	<i>ureC</i>	Urease subunit alpha (α); EC 3.5.1.5; dinuclear-Ni active site	567	Cytoplasm
PP_2846	<i>ureE</i>	Urease accessory protein UreE (Ni metallochaperone)	167	Cytoplasm
PP_2847	<i>ureJ</i>	Urease accessory protein UreJ (integral membrane, ~5 TM helices)	181	Membrane
PP_2848	<i>ureF</i>	Urease accessory protein UreF	224	Cytoplasm
PP_2849	<i>ureG</i>	Urease accessory protein UreG (SIMIBI GTPase)	207	Cytoplasm

This confirms that *P. putida* encodes all three structural subunits (UreABC) and all four canonical maturation accessory proteins (UreD/UreE/UreF/UreG), with *ureF* co-transcribed adjacent to its direct partners *ureE/ureJ* and *ureG*. The additional membrane protein **UreJ** is characteristic of *Pseudomonas*-type *ure* clusters and is distinct from UreF's soluble role.

Biochemical pathway. UreF operates in **urease metallocenter biosynthesis**, the maturation branch of urea catabolism. The physiological output of the pathway is an active urease that hydrolyzes urea $\rightarrow 2 \text{ NH}_3 + \text{CO}_2$, allowing the cell to assimilate urea-derived nitrogen (and, secondarily, to buffer against acid/produce ammonia); bacterial urease expression is typically nitrogen-regulated, urea-inducible, or constitutive [PMID:7565414]. UreF is **not** part of urea transport or of downstream ammonia assimilation; its role is strictly upstream, in producing the catalytically competent enzyme.

5. Evidence summary

- **Experimental (orthologs, high-confidence, directly transferable by strong homology):** crystal structures of UreF, UreF–UreH, and UreG/UreF/UreH [PMID:20635345; 22013070; 24115911]; site-directed mutagenesis defining UreF's UreG-binding surface and essential residues Tyr48/Arg250/Lys165 [PMID:22013070; 22369361]; in vitro reconstitution of GTP-dependent, Ni-charged UreG dimer formation and nickel release [PMID:24115911]; UreF control of nickel-exclusion/activation fidelity [PMID:8808930]; nickel-tunnel disruption reducing Ni incorporation [PMID:26401965]; foundational reconstitution of the UreD–UreF–UreG–apourease activation complex [PMID:10500143; 8808930; 9209019].
- **Evolutionary (target-specific homology):** Global (Needleman–Wunsch) alignment of the *P. putida* UreF (Q88J01, 224 aa) shows **55.4% amino-acid identity** to the biochemically characterized *K. aerogenes* UreF (P18318, also 224 aa) and 25.9% identity to *H. pylori* UreF (Q9S0Q2). The high identity and identical length to the Gammaproteobacterial *K. aerogenes* ortholog strongly justify transferring its structural/mechanistic conclusions to the *P. putida* protein. At the residue level, the fidelity-linked *K. aerogenes* **Lys165** (K165A uncouples UreG GTP hydrolysis from urease activation; [PMID:22369361]) is **conserved as Lys165** in the *P. putida* protein, and the C-terminal UreG-recruitment region ([PMID:22013070]) is highly conserved (~63% identity over the last ~70 aligned residues) — indicating the target retains the same functional surfaces.
- **Database/inference (target-specific):** UniProt Q88J01 functional, subunit, and localization annotations (HAMAP MF_01385); genomic cluster mapping (PP_2842–PP_2849); UreF-family domain assignment (PF01730 / IPR002639 / IPR038277).
- **Caveat:** No study focuses on the *P. putida* KT2440 UreF protein specifically; its assignment rests on strong sequence/family homology to the biochemically and structurally

characterized enterobacterial and *H. pylori* orthologs plus curated computational annotation.

6. Supported and refuted hypotheses

Supported - UreF is a cytoplasmic urease-maturation chaperone, not a catalytic enzyme. (UniProt Q88J01; [PMID:23539618; 7565414]) - UreF scaffolds UreD/UreH and recruits/activates the UreG GTPase to form the pre-activation complex. ([PMID:22013070; 22369361; 24115911]) - UreF participates in gated, selective nickel delivery (tunnel + fidelity control), preventing premature/mismetalated active sites. ([PMID:26401965; 8808930]) - In *P. putida* KT2440, *ureF* is embedded in a complete *ure* cluster and matures the co-encoded UreC apoenzyme. (genome mapping)

Refuted / excluded - UreF catalyzes urea hydrolysis — refuted (catalysis is by UreABC/UreC; UreF has no EC number and no substrate). - UreF is a nickel metallochaperone that itself binds/delivers nickel as the primary carrier — not supported; nickel binding/delivery is attributed to UreE→UreG, while UreF's role is to build/activate the UreG donor and form part of the transfer conduit ([PMID:24115911; 26401965]). - UreF is membrane-associated — refuted for UreF (cytoplasmic); the membrane component of the *P. putida* cluster is UreJ (PP_2847), a different protein.

7. Limitations and future directions

- Direct biochemistry on the *P. putida* UreF is lacking; conclusions rely on homology to *K. aerogenes* and *H. pylori*. A dedicated *P. putida* knockout/complementation and in vitro activation study would confirm transferability.
 - The exact function of the *Pseudomonas*-specific accessory protein **UreJ** (membrane) and whether it modulates UreF/UreG activity or urea/nickel access is unresolved and worth investigating.
 - The precise coupling between UreG GTP hydrolysis and UreF-mediated conformational gating (the "GAP-like" step) remains mechanistically debated [PMID:22369361] and could be probed by pre-steady-state kinetics on the *Pseudomonas* complex.
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Key references

- P 24115911
— UreG/UreF/UreH complex structure and maturation mechanism (definitive).
- P 22013070
— UreF–UreH structure; UreG recruitment (Tyr48/Arg250).
- P 20635345
— First UreF structure (all- α fold, dimer).
- P 22369361
— UreF UreG-binding site; activation fidelity/GAP hypothesis.
- P 8808930
— UreD–UreF–urease complexes; nickel-exclusion/fidelity.
- P 26401965
— Nickel-transfer tunnel through UreD/UreF.
- P 35556134
— Review: nickel trafficking along the urease maturation pathway.
- P 23539618 /
20046957 / 7565414 — Reviews of urease metallocenter biosynthesis and molecular biology.
- UniProt Q88J01 — target-specific curated annotation.